

Exam Tips for TTTE6104 Software Engineering Paradigm

Semester 1 2025/2026

Exam date, time and venue: 29/01/2026 (Thursday), 9.00am @ BK2

Reading Materials

Ensure you have a solid grasp of the following topics and their associated principles:

1. **Requirement Engineering and Software Design**
2. **Software Construction and Software Testing**
3. **Software Maintenance**
4. **Object-Oriented Software Engineering: Tools and Practices**
5. **Agile Methodology: Tools and Practices**

Exam Format

- **2 Main Questions** with sub-questions.
- **Open-Ended** format.
- Each question is worth **30 marks** (Total: 60 marks).
- Exam Duration: **2 hours**.

Key Preparation Tips

1. **Understand Core Concepts**
 - Revise the foundational principles of software engineering paradigms, such as **Agile, Waterfall, V-Model, Spiral, and DevOps**.
 - Pay attention to how paradigms differ from one to another.
 - Be prepared to critique paradigms and suggest the most appropriate one for specific **project types, team sizes, or constraints**.
2. **Role of Stakeholders**
 - Review the interests and expectations of different groups of individuals (stakeholders) in a software engineering project.

- Understand how different paradigms impact stakeholders such as **developers, clients, end-users, and project managers.**

3. **Real-World Applications**

- Learn about the practical applications, strengths, and challenges of each paradigm in actual projects.
- Strategies to mitigate the challenges.
- Recap your presentation on issues and challenges.

4. **Comparison and Analysis**

- Expect questions that ask for comparisons among paradigms. Focus on aspects like:
 - Suitability for small vs. large projects.
 - Adaptability to changing requirements.
- Practice analysing these factors with clarity and supporting your answer with **examples.**

5. **Agile Methodology and Tools**

- Familiarize yourself with Agile practices you have learned from class.
- Understand the role of refactoring in Agile development.

6. **Practice Analytical and Writing Skills**

- Practice writing **concise, well-structured answers** using examples to illustrate your points.

7. **Use Case Studies**

- Incorporate real-world examples or case studies to show your understanding of how paradigms are applied in various scenarios.